Paper #102: Substrate Computation — Computing with Vacuum Eigenvalues

The Casimir effect IS computation. The proton IS a register.

May 4, 2026

Substrate Computation: Computing with Vacuum Eigenvalues

Physics IS computation on D_IV^5. The Casimir effect is a spectral evaluation. The proton stores 42 bits. The mass gap is an 8-state register.

Abstract

We propose that every physical process is a computation on the eigenvalue ladder of D_IV^5 = SO_0(5,2)/[SO(5) x SO(2)]. The Casimir effect between parallel plates IS a spectral evaluation: plates set boundary conditions (the program), the vacuum evaluates the truncated eigenvalue sum (the computation), and the resulting force is the output. The proton stores C_2g *= 42 bits of information in its Hamming(7,4,3) error-corrected spectral recording. The mass gap at lambda_1 = C_2 = 6 has d(1)+1 = g+1 = rank^3 = 8 distinguishable states — one byte. The fine structure constant alpha = 1/N_max = 1/137 is the inference cost: each measurement consumes 1/N_max of the total spectral capacity. The total spectral parallelism of D_IV^5 is sum P(k) for k=0..9 = 8008 = rank^3*gc_2*c_3 channels — expressible entirely in BST integers and Chern classes. We describe six computational primitives (CREATE, ANNIHILATE, SHIFT, BRANCH, MERGE, MEASURE) that operate at femtometer scale, and six device designs from near-term (Hamming memory cell, Fibonacci interconnect) to far-future (substrate recorder). The Casimir harvester at N_max = 137 planes powers the substrate computer with efficiency bound eta = n_C/g = 5/7 = 71.4%. The universe has been computing in bytes for 13.8 billion years. We propose to join the computation.

Section 1: The Computational Model

Three levels: spectral (substrate itself), topological (error-corrected), classical (our technology). Physics IS the output of Level 1.

Section 2: The Heptit — BST's Computational Primitive

d(1) = g = 7 states. d(1)+1 = rank^3 = 8 = byte. Shannon capacity C(1) = g/rank = 3.5 bits. 11 gate generators from SO(5) x SO(2). Gate space dim = d(3) = 77 = g*11. Native Hamming(7,4,3). Gate depth per EC = N_*cn_C = 15.

Section 3: Casimir as Computation (T1686)

Plates = program. Force = output. Every Casimir measurement IS a spectral evaluation. The vacuum computes the eigenvalue sum in real time. Programmable Casimir array = spectral computer.

Section 4: The Proton Register (T1684)

42 = C_2*g Hamming bits. Proton = minimum stable recording. C_2 error-correction slots, g modes each. Windings per macro-bit = rank^2*(rank*n_C+1) = 44. Proton lifetime > 10^34 years = code never breaks.

Section 5: Information Density

Bekenstein at charge radius: 36 bits. BST Hamming: 42 bits. Factor g/C_2 = 7/6. Nuclear information density: ~10^45 bits/m^3. All human data ever: ~10^22 bits. One cubic meter of nuclear matter = 10^23 times all human information.

Section 6: Six Computational Primitives

CREATE (lambda_0 → lambda_k), ANNIHILATE (lambda_k → lambda_0), SHIFT (lambda_k → lambda_{k+1}), BRANCH (lambda_k → lambda_a + lambda_b), MERGE (lambda_a + lambda_b → lambda_k), MEASURE (project to |lambda_k>). Particle physics IS substrate computation.

Section 7: Device Designs

Eigenvalue register (7 NV centers at Hamming spacing), spectral ALU (geodesic multiplication), Hamming memory cell (7 qubits), topological bus (Cheeger-protected), Fibonacci interconnect (golden angle routing), vacuum logic gate (MEMS plates at BST gaps).

Section 8: Communication via Substrate

Entangled pairs at k=1 eigenvalue = optimal channel. Discrete series = Hamming-protected = proton lifetime coherence. Bandwidth: N_max = 137 independent channels. Substrate communication = quantum teleportation at the optimal spectral address.

Section 9: Energy Source — Casimir Harvester

Gap: N_max * a_BaTiO3 = 54.9 nm. Efficiency: n_C/g = 5/7. Switching: epsilon_ratio = n_C = 5 (BaTiO3). Power: THz cycling at BST resonance. The vacuum powers the computer.

Section 10: Implications

The universe computes. Physics = computation output. Matter = recorded data. Consciousness = decoding. The five BST integers = the instruction set. Alpha = 1/137 = the overhead. We don't build the computer. We learn to program the one that's already running.

8008 channels. 42 bits per proton. 1/137 overhead per measurement. The substrate has been computing for 13.8 billion years.